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Electronic White Cane with GPS Radar-Based Concept as Blind Mobility Enhancement without Distance Limitation

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Abstract. The Indonesian Ministry of Social Affairs, in July 2012, informed that the number of blind in Indonesia has been the largest among to the people with other disabilities. The most common tools utilized to help the blind was a conventional cane which has limited features and therefore it was difficult to be used as a mobilization tools. Moreover, the conventional cane cannot assist them or their family when the blind gets lost. In this research, we designed and implemented an electronic white cane with the concept of radar and global positioning system (GPS). The purpose of this research was to design and develop an electronic white cane which can enhance the mobility of the blind without distance coverage limitation. Utilizing ultrasonic sensors as a distance measurement and a servo motor as an actuator, the produced radar system is able to map an area with maximum distance and coverage angle of 5 meters and 180^0 respectively. The blind senses the obstacle around them from the vibration generated by five vibration motors. The vibration becomes more intense when the obstacle is detected closer. In addition, we implemented a GPS to monitor the blind's position and allow their family to find them easily when the blind need a help. Based on the tests performed, we have successfully developed an electronic white cane that can be a solution to improve the blind's mobility.

23 July 2026 19:00:34

INTRODUCTION

The World Health Organization (WHO) in 2010 estimated that more than 285 million people out of 6,697 million world population were affected by visual impairments [1]. They approximate 90% of visually impaired live in developing countries including Indonesia [2]. The Indonesian Ministry of Social Affairs in July 2012 informed that 1,749,981 people are suffering blind. The number of blind in Indonesia was the largest compared to the people with other disabilities [3]. The number was predicted will continuously grow due to the increasing number of elderly population in Indonesia. At the same time, more people will be at risk of visual impairment particularly blind due to chronic eye diseases, unhealthy live style and ageing processes [3].

In order to improve the blind mobility, the most common tool utilized to help their mobility is a conventional cane which has limited features. The tool gives responses to the blind when touching an object which are limited in only one direction and the distance is less than two meters. Moreover, blind is facing another problem when they are get lost. There are no devices implemented to assist or to guide their family when the blind is get lost. In this research, we developed an electronic white cane equipped with electronics sensor and an artificial intelligent to enhance blind mobility. For that purpose, we designed and implemented radar system that are able to map surrounding area with maximum distance and coverage angle of 5 meter and 180^0 respectively. The blind sense the surrounding condition from the vibration generated by motors. In additional, we also implement a global positioning system (GPS) to monitor the blind's position. Once the blind gets lost, their family can easily find their position using map installed in smart phone.

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METHODHOLOGY

Design and development of the electronic white cane

We designed and developed an electronic white cane utilizing three ultrasonic sensors and a servo motor as a radar system to map an area in front of the cane. Ultrasonic sensors were employed as distance measurement while a servo motor was used as an actuator to scan the area. In addition, a GPS system was installed in the cane in order to accurately informed the position of the cane to a formerly programmed smart phone.

Ultrasonic sensing techniques are well known and are widely used in various fields of engineering both for industrial and medical applications. The advantage of using ultrasonic sensing was its outstanding capability to measure object nondestructively. Ultrasonic waves can propagate through any kinds of media including solids, liquids and gases [4]. The sensor detects objects by emitting a short ultrasonic burst with frequency more than 20,000 Hz and then listening for the echo. Figure 1 show the basic principle of an ultrasonic sensor operation. The sensor provides an output pulse as short burst to the object and will be terminated when the echo burst is detected. The time width of the pulse corresponds to the distance of the target [5]. The effect of the air temperature around the sensor should be take into account in order to accurately measure the distance based on the following equation:

$$S = \frac{C_{air} x t}{2} \tag{1}$$

$$S = \frac{\{331 + (0.6 x T_c)\}t}{2} \tag{2}$$

where, **S** is the measured distance, **t** is time delay, **C_{air}** is ultrasonic velocity on the air, and **T_c** is temperature of the air [5]. In order to measure a distance, an ultrasonic sensor is controlled by a microcontroller. First, microcontroller send an information by setting a pin into high logic and therefore an ultrasonic burst is generated. At that moment, internal clock is counting up until ultrasonic echo is received. The time delay between first burst and echo will be converted into actual distance and is stored inside microcontroller memory.

23 July 2026 19:00:34

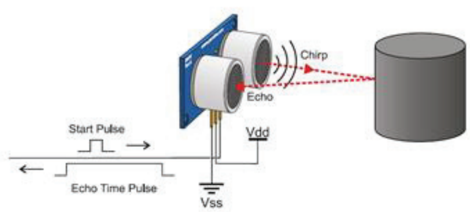

FIGURE 1. Basic operation of distance measurement using ultrasonic sensor [5]

Ultrasonic sensors can only detect an object which is located in the front of the sensors within a certain angle range. In order to enhance the coverage of the measurement area, a servo motor is used to rotate the ultrasonic sensors. This technique is similar to the radar system. Radar is an object detection system which uses radio waves to determine the range, altitude and direction of objects. The radar transmits pulses of radio waves which bounce off any object in their path [6]. To rotate ultrasonic sensors, a servo motor is controlled by sending a pulse width modulation (PWM). In this research, the motor is controlled to move the ultrasonic sensors 80 degrees either clockwise or counter-clockwise. The PWM sent to servo motor determines position of the shaft. Figure 2 shows the relation between shaft position and length of the PWM.

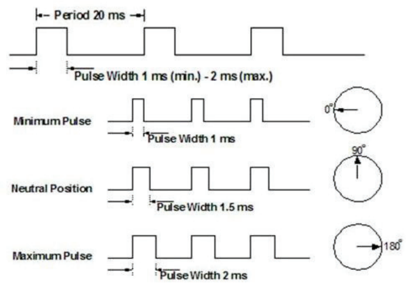

FIGURE 2. Servo Shaft Position and Length of PWM [7]

Figure 3 shows the illustration of the radar system consisting of shaft motor servo and ultrasonic sensor in detecting the object in the area within 180^0 in front of the white cane.

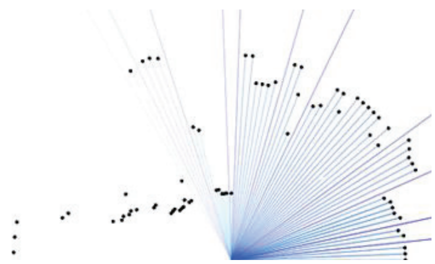

FIGURE 3. Illustration of the object detection system using a Radar based motor servo and ultrasonic sensor

Next, we designed and developed an electronic vest that will directly inform the existence of obstacle to the blind. For that purpose, five vibration motors were installed in the vest. The vibration intensities were controlled by Arduino microcontroller informing the distance between white cane and the object. Shorter the object to the white cane, more intense the vibration will be. Bluetooth module was employed to communicate between the cane and the vest.

Since the blind are susceptible for getting lost, a Global Positioning System (GPS) device was installed to the electronic white cane. GPS is a system that can be used to inform an object position globally in the Earth's surface using satellites. A unique signal and orbital parameters related to the object position are transmitted by satellites. GPS devices will decode and compute the precise location of the object based on the information received from the satellites [8]. The type of GPS device used in this work was A-GPS. This GPS relies on a server to acquire location. This feature comes useful in an environment where GPS takes longer time to establish connectivity with 4 satellites [9]. GPS tracker will receive satellite and server signals and provide an additional information, helping on location calculations. Later on, server will forward the information to the GPS user. In our system, the information of the blind location was sent by pushing a switch button. Using General Packet Radio Service (GPRS) signal equipped in the GPS, the family can track the location of the blind. Acquired position is then translated as data information and sent through SMS or web tracking with the help of GPRS signal. Data information was formatted in the form of Google Maps link that can be accessed through browser easily. Using GPS tracker, users can find coordinate location of the implanted GPS device and monitor the position through Android smartphone.

23 July 2026 19:00:34

RESULTS AND DISCUSSION

We have successfully developed an electronic cane with GPS radar-based concept consist of 2 main parts, an electronic white cane which can detect objects with area coverage 180^0 and electronic vest to inform blind's surrounding situation through motors vibration. The electronics vest is also used to inform the blind's position to their families through GPRS signal. The electronic white cane and electronic vest are connected through Bluetooth device. Figure 4 shows the developed electronics white cane and the electronics vest

(a) (b)

FIGURE 4. Electronic white cane (a) and electronics vest (b)

23 July 2026 19:00:34

The electronics white cane consists of five main components; three ultrasonic sensors, a servo motor, a microcontroller, a Bluetooth module and five vibration motors. As shown in Figure 4, the three ultrasonic sensors are located at three different heights and are connected to a servo shaft in order to rotate the direction in 180 degrees. The application of three different level of sensor height gives an advantage to detect an object in the range of 3 cm to 120 cm height. By using a microcontroller Arduino Nano and a servo motor, the direction of ultrasonic sensors can be controlled precisely.

Figure 5 shows the simplified communication system of each components. By pushing the on push-button, the microcontroller sends a PWM signal to servo motor to change the direction of ultrasonic sensors into specified direction. First, a servo motor locates the ultrasonic sensors at 0^0 from the front of the cane. The three ultrasonic sensors measure the distance of the object in that specified angle. The servo motor then rotates the ultrasonic sensor direction to 45^0, 90^0, 135^0 and 180^0 sequentially. After collecting all the surrounding object distance, servo goes to default angle and the microcontroller sends the mapping information thought Bluetooth device to the vest. The electronics vest receives the surrounding object distance and convert the information into vibration. The blind will sense the surrounding condition through the vibration generated by five motors. When an obstacle approaches the sensor, the vibration becomes more intense and vice versa. Finally, the blind is able to identify and analyze the surrounding without tipping every single object. Figure 6 shows the flow chart of the system.

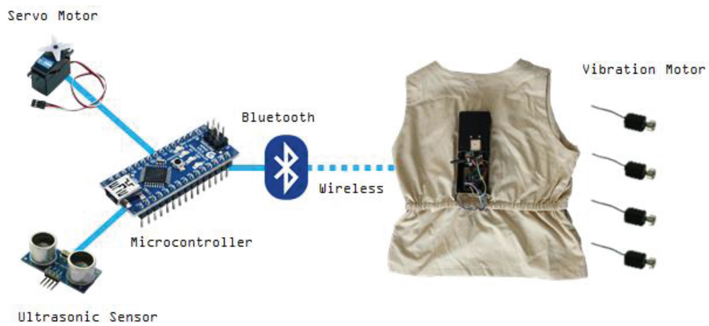

FIGURE 5. Electronics Stick Interconnecting Components

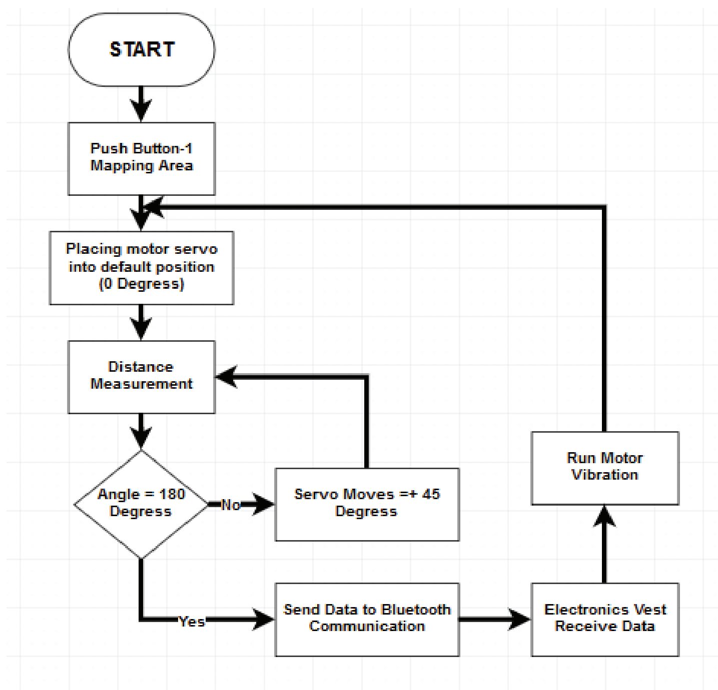

FIGURE 6. The Flow Chart of Blind's Mapping Area

Blind coordinate position

Due to their disabilities to recognize the position, blinds are facing a potential to get lost specially when they are entering a new location. It is difficult for the blind to inform their exact location to their family. By applying the GPS module to the vest, the blind can immediately inform their location to their family. This was done by pushing a single push-button located in the cane. Once the button is pushed, the microcontroller sends a signal into GPS module and the GPS module starts to estimate blind's position. Using relative position from several satellites, the GPS module knows location on the Earth surface. After receiving actual position based on latitude and longitude coordinate, GPS module sends the information back to Arduino. The Arduino microcontroller forwards the coordinate position to the blind's families through the internet communication by using GSM module. Blind's family will receive a notification on their smartphone. In short, the blind's family can easily find the blind position. Using our developed application in the smart phone, the blind's family can find the blind position instantly. Figure 7 and Figure 8 shows the mechanism to find blind position and the developed Android Application to find the blind's position respectively.

23 July 2026 19:00:34

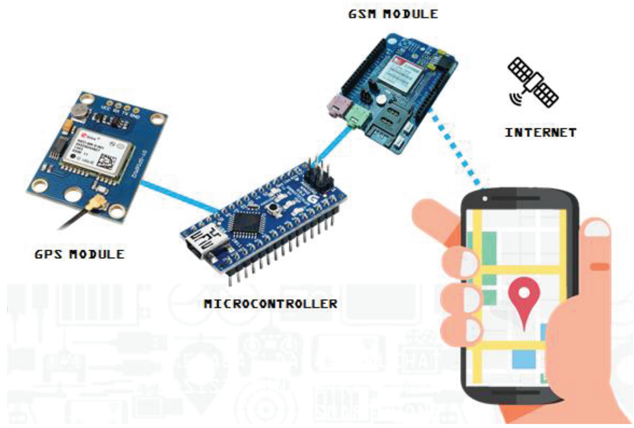

FIGURE 7. The Flow Chart of Blind's Mapping Area

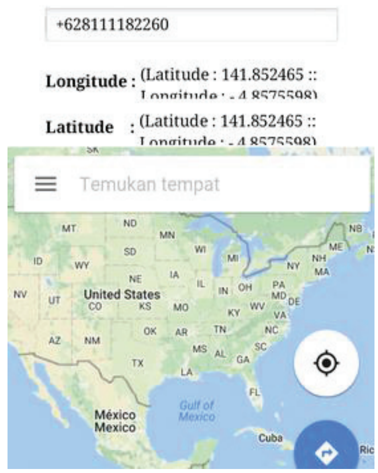

FIGURE 8. Blind's position monitoring application on Smartphone

The developed white cane has been implemented to actual blind people, and it was found that the device can be an alternative solution to improve the blind's mobility

CONCLUSION

We have successfully developed and implemented the electronic white cane equipped with GPS and radar-based concept. The white cane has ability to map an area within 5-meter distance and 180^0 angle. The blind senses the condition around them from the vibration generated by five vibration motors. The intensity of the vibration reflects the distance of the object from the white cane position. Using the developed device, the blind was able to identify and analyze the surrounding without tipping every single object. Moreover, blind's position can be monitored from distance using a global positioning system (GPS). Blind's family will receive notification on their smartphone when blind feels lost or needs help. Based the tests performed on actual blind people, the device can be an alternative solution to improve the blind's mobility.

23 July 2026 19:00:34

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23 July 2026 19:00:34